Computers & Fluids 105 (2014) 58–65

Contents lists available at ScienceDirect

Computers & Fluids

journal homepage: www.elsevier.com/locate/compfluid

A nodal discontinuous Galerkin lattice Boltzmann method for fluid flow problems

A. Zadehgol, M. Ashrafizaadeh, S.H. Musavi *

Department of Mechanical Engineering, Isfahan University of Technology, Isfahan 8415683111, Iran

ARTICLE INFO

Article history:
Received 20 January 2014
Received in revised form 24 June 2014
Accepted 3 September 2014
Available online 16 September 2014

Keywords:
Lattice Boltzmann
Nodal discontinuous Galerkin
Fluid flow

ABSTRACT

We present a new application of the nodal discontinuous Galerkin method in solving the lattice Boltzmann equation. By splitting the collision and the streaming operators of the lattice Boltzmann equation, the nodal discontinuous Galerkin method is implemented to solve the resulting advection equation. Space discretization is performed using unstructured grids with triangular elements, while time marching is carried out by applying the low-storage fourth-order, five-stage Runge–Kutta method. To validate the results, two well-known benchmark problems are simulated. The lid-driven square cavity at Reynolds numbers of 1000 and 5000 and Mach number of 0.1 and the impulsively started cylinder at Reynolds numbers of 550 and 9500 with Mach numbers of 0.1 and 0.05, respectively. The results show excellent agreement between the present study and the existing results in the literature. The method is then used to simulate flow in a porous medium. It is shown that the present method requires significantly lower number of grid points compared to the standard lattice Boltzmann method for the simulation of the porous medium flow.

1. Introduction

The lattice Boltzmann method (LBM) has been widely used for simulating fluid flow problems, in the past decades [1,2]. Historically, the lattice Boltzmann method is derived from the lattice gas automata [3]. Later on, it was shown first by McNamara et al. [4] and then by Sterling and Chen [5] and He and Luo [6] that the lattice Boltzmann equation is a special discretization of the Boltzmann equation. Therefore, the velocity space and the physical space can be decoupled. Since then different numerical schemes, such as the finite difference (FD-LBM) [7–11], finite volume (FV-LBM) [12–15], and finite element (FE-LBM) [16–18], have been utilized to discretize and solve the Boltzmann equation.

In the finite difference method a computational molecule is replicated throughout the domain which requires only local grid data. While this approach simplifies the calculations, it becomes inconvenient for the curved boundary treatments. In the finite volume method, the space is broken into several finite volumes, and a cell averaged value is considered for the entire finite volume, while the flux values on the boundaries are calculated based on some approximations, such as the central or the Lax-Friedrichs schemes. While this method is flexible for the treatment of the complex geometries, the cell averaging inside each element prevents the possibility of achieving higher order accuracies. To overcome the above mentioned shortcoming of the finite volume method, the natural solution is to increase the number of nodes inside each element. This approach is the basis of the more advanced finite element method, which is considered as a higher order approach in the sense that it enables one to utilize a higher order interpolation function. On the other hand, the globally defined basis function and the requirement that the same test functions should be orthogonal to the residuals, introduce an inconvenience since the semi-discrete form becomes implicit and the resulting global mass matrix should be inverted [19]. The discontinuous Galerkin (DG) method has been proposed aiming to benefit from the best features of the finite volume and the finite element methods, since it introduces the concept of the numerical flux of the finite volume method into the classical finite element method. In the discontinuous Galerkin method the field variables are allowed to have discontinuous values on the interface of any two neighboring elements. The discontinuity is to be resolved by using the numerical flux.

Discontinuous Galerkin spectral element method based on triangular element was used by Shi et al. [20] to solve the lattice Boltzmann equation in the discrete velocity space. They chose Dubiners modified triangular basis [21] for the test function and

* Corresponding author. Tel.: +98 311 3913919; fax: +98 311 3912628.
E-mail addresses: a.zadehgol@me.iut.ac.ir (A. Zadehgol), mahhud@cc.iut.ac.ir (M. Ashrafizaadeh), hmusavi@me.iut.ac.ir (S.H. Musavi).

http://dx.doi.org/10.1016/j.compfluid.2014.09.015
0045-7930/

simulated flow past a circular cylinder at relatively low Reynolds numbers ($Re \leqslant 200$). Duster et al. [22] implemented the discontinuous Galerkin lattice Boltzmann method with quadrilateral elements and applied the shape functions proposed by Szabo and Babuška [23]. They used a first-order forward Euler scheme for the time discretization. Min and Lee [24] used the spectral-element discontinuous Galerkin lattice Boltzmann method which utilizes quadrilateral element based on Gauss–Lobatto–Legendre grids and tensor product basis of the one-dimensional Legendre-Lagrange interpolation polynomials. Decoupling the collision step from the streaming step they could achieve numerical stability at high Reynolds numbers. Although the two dimensional mass matrix in their method becomes diagonal, their method is restricted to quadrilateral elements.

In this work we present a new application of the nodal discontinuous Galerkin method, which was proposed by Hesthaven and Warburton [19], for solving the lattice Boltzmann equation. The collision and the streaming operators of the lattice Boltzmann equation have been split, and the resulting advection equation is solved using the nodal discontinuous Galerkin method. The Jacobi orthonormalized polynomials for the standard triangle are used as the basis function. Although the element mass matrix in the present method is not diagonal, this is not as important because the inverse of the element mass matrix is obtained only at the beginning of the computations and used for all the time steps. The time marching is performed by applying the low-storage fourth-order, five-stage Runge–Kutta method (LSERK). Application of triangular elements and the operator splitting technique in solving the lattice Boltzmann equations with the discontinuous Galerkin method, has made our scheme capable of simulating fluid flow problems in complex geometries and at relatively high Reynolds numbers.

The paper continues with a brief description of the formulation and numerical features of the method. To validate the numerical scheme, two well-known benchmark fluid flow problems are simulated; first the lid-driven square cavity at Reynolds numbers of 1000 and 5000 and a Mach number of 0.1, and second the impulsively started cylinder at Reynolds numbers of 550 and 9500 with Mach numbers of 0.1 and 0.05, respectively. In each case the results are compared with those of other numerical and/or experimental data in the literature. Next, the method is used to evaluate the permeability of a porous medium. In this case, the performance of the presented method is compared with that of the standard LBM in terms of required number of grid points. The paper ends with some concluding remarks.

2. Formulation and numerical procedure

In this section, first the formulation of the lattice Boltzmann method is given and then the nodal discontinuous Galerkin numerical procedure for treating the lattice Boltzmann equation is explained.

2.1. Lattice Boltzmann equation

The discrete Boltzmann equation with the so called BGK collision approximation [25] reads:

$$\frac{\partial f_i}{\partial t} + c_{i,\alpha}\frac{\partial f_i}{\partial x_\alpha} = -\frac{1}{\lambda}(f_i - f_i^{eq}), \; i = 0,1,\ldots,b, \tag{1}$$

where f_i is the velocity distribution function along the discrete particle velocity $\mathbf{c}_i$, f_i^{eq} is the equilibrium distribution along the same direction i, λ is the relaxation time towards equilibrium, and b is the total number of discrete particle velocities. In this study we simulate two-dimensional cases with the D2Q9 lattice given as

$$\begin{aligned} &\mathbf{c}_0 = \mathbf{0}, \\ &\mathbf{c}_i = \cos(i-1)\frac{\pi}{4}\mathbf{e}_x + \sin(i-1)\frac{\pi}{4}\mathbf{e}_y, \; i = 1,3,5,7, \\ &\mathbf{c}_i = \sqrt{2}\left[\cos(i-1)\frac{\pi}{4}\mathbf{e}_x + \sin(i-1)\frac{\pi}{4}\mathbf{e}_y\right], \; i = 2,4,6,8, \end{aligned} \tag{2}$$

where $\mathbf{e}_x$ and $\mathbf{e}_y$ are the unit vectors along x and y directions, respectively, and the equilibrium distribution function defined as

$$f_i^{eq} = \rho t_i\left(1 + \frac{c_{i,\alpha}u_\alpha}{c_s^2} + \frac{(c_{i,\alpha}u_\alpha)^2}{2c_s^4} - \frac{u_\alpha^2}{2c_s^2}\right), \tag{3}$$

where ρ and u_α are the (macroscopic) density and velocity, respectively, $c_s^2 = 1/3$ is the squared sound speed of the lattice, and t_i is the weighting factor for the direction i, where $t_0 = 4/9, t_{1,3,5,7} = 1/9$, and $t_{2,4,6,8} = 1/36$.

By formally integrating Eq. (1) over the time step δt we arrive at

$$f_i(x_\alpha + c_{i,\alpha}\delta t, t+\delta t) - f_i(x_\alpha, t) = -\frac{1}{\lambda}\int_t^{t+\delta t}(f_i - f_i^{eq})dt'. \tag{4}$$

By using the trapezoidal rule for the integral of the right hand side of Eq. (4) we obtain

$$\begin{aligned} &f_i(x_\alpha + c_{i,\alpha}\delta t, t+\delta t) - f_i(x_\alpha, t) \\ &\quad = -\frac{1}{2\tau}\left[(f_i - f_i^{eq})|_{(x_\alpha + c_{i,\alpha}\delta t, t+\delta t)} + (f_i - f_i^{eq})|_{(x_\alpha, t)}\right], \end{aligned} \tag{5}$$

where $\tau = \lambda/\delta t$ is the dimensionless relaxation time. In order to transform the implicitly coupled Eq. (5) into an explicit lattice Boltzmann equation, a modified velocity distribution function $\bar{f}_i$ and its corresponding equilibrium distribution $\bar{f}_i^{eq}$ have been introduced as [26]

$$\bar{f}_i = f_i + \frac{1}{2\tau}(f_i - f_i^{eq}), \text{ and } \bar{f}_i^{eq} = f_i^{eq}, \tag{6}$$

which results in the lattice Boltzmann equation of the form:

$$\bar{f}_i(x_\alpha + c_{i,\alpha}\delta t, t+\delta t) - \bar{f}_i(x_\alpha, t) = -\frac{1}{\tau + 1/2}(\bar{f}_i - \bar{f}_i^{eq})|_{(x_\alpha, t)}. \tag{7}$$

It is common in the literature to solve the lattice Boltzmann Eq. (7) in two separate steps, namely, collision:

$$\tilde{f}_i(x_\alpha, t) = \bar{f}_i(x_\alpha, t) - \frac{1}{\tau + 1/2}(\bar{f}_i - \bar{f}_i^{eq})|_{(x_\alpha, t)}, \tag{8}$$

and streaming:

$$\bar{f}_i(x_\alpha + c_{i,\alpha}\delta t, t+\delta t) = \tilde{f}_i(x_\alpha, t). \tag{9}$$

Although Eq. (9) is the exact solution of the pure advection equation, it restrains the numerical solution to uniform mesh and unit CFL number which is the natural result of coupling the velocity space with the physical space. An elegant way to overcome these deficiencies of the standard lattice Boltzmann method is to come back to the pure advection equation for the streaming process, that is

$$\frac{\partial \bar{f}_i}{\partial t} + c_{i,\alpha}\frac{\partial \bar{f}_i}{\partial x_\alpha} = 0. \tag{10}$$

There are some numerical techniques to treat Eq. (10). In this study we use the nodal discontinuous Galerkin scheme, to be described in the next section, to discretize this equation.

The macroscopic properties, density and velocity can be related to the moments of the distribution function as follows:

$$\rho = \sum_{i=0}^{b}\bar{f}_i, \tag{11}$$

$$\rho u_\alpha = \sum_{i=0}^{b}c_{i,\alpha}\bar{f}_i. \tag{12}$$

It can be shown by the Chapman-Enskog asymptotic analysis that the kinematic viscosity is related to the dimensionless relaxation time as [27]

$$\nu = \tau c_s^2 \delta t. \tag{13}$$

2.2. Nodal discontinuous Galerkin method

In this section we explain the nodal discontinuous Galerkin method proposed by Hesthaven and Warburton [19] and the application of this method in solving the streaming Eq. (10). By introducing the flux $F_{i,\alpha}(\bar f) = c_{i,\alpha}\bar f_i$, we rewrite Eq. (10) as follows

$$\frac{\partial \bar f_i}{\partial t} + \frac{\partial F_{i,\alpha}(\bar f)}{\partial x_\alpha} = 0. \tag{14}$$

As in all of the finite element methods, the problem domain Ω is partitioned into some no-overlapping sub-domains Ω^k called elements such that $\bigcup_{k=1}^{K}\Omega^k = \Omega$, where K is the total number of elements. The local weak form of Eq. (14) can be derived by taking its inner product to a local test function ϕ over the considered element Ω^k equal to zero:

$$\int_{\Omega^k}\left(\frac{\partial \bar f_i}{\partial t} + \frac{\partial F_{i,\alpha}(\bar f)}{\partial x_\alpha}\right)\phi d\Omega = 0. \tag{15}$$

Using integration by parts for the flux term, we have

$$\int_{\Omega^k}\frac{\partial \bar f_i}{\partial t}\phi d\Omega - \int_{\Omega^k}F_{i,\alpha}(\bar f)\frac{\partial \phi}{\partial x_\alpha}d\Omega = -\int_{\Gamma^k}F_{i,\alpha}(\bar f)n_\alpha \phi d\Gamma, \tag{16}$$

where Γ^k represents the boundary of the element Ω^k, and n_α is the unit normal vector of Γ^k. In the discontinuous Galerkin scheme, the values of the field variable $\bar f_i$ on the interface between two neighboring elements can be discontinuous, i.e. we have $\bar f_i^-$ for the considered element and $\bar f_i^+$ for the neighboring element. The numerical flux $F^*_{i,\alpha} = F^*_{i,\alpha}(\bar f^-, \bar f^+)$ is defined on the elements boundary to resolve the discontinuity. This numerical flux is introduced in the formulation by replacing the analytical flux on the boundary (right hand side of Eq. (16)) to obtain

$$\int_{\Omega^k}\frac{\partial \bar f_i}{\partial t}\phi d\Omega - \int_{\Omega^k}F_{i,\alpha}(\bar f)\frac{\partial \phi}{\partial x_\alpha}d\Omega = -\int_{\Gamma^k}F^*_{i,\alpha}(\bar f)n_\alpha \phi d\Gamma. \tag{17}$$

Integrating by parts of Eq. (17), we arrive at the more useful local weak form

$$\int_{\Omega^k}\left(\frac{\partial \bar f_i}{\partial t} + \frac{\partial F_{i,\alpha}(\bar f)}{\partial x_\alpha}\right)\phi d\Omega = \int_{\Gamma^k}[F_{i,\alpha}(\bar f) - F^*_{i,\alpha}(\bar f)]n_\alpha \phi d\Gamma. \tag{18}$$

This equation is used as the basis for treating the streaming process in this study. For the numerical flux, several schemes have been proposed in the literature, among them, the Lax-Friedrichs flux which takes into account the upwinding effects, is chosen. The Lax-Friedrichs flux reads

$$F^*_{i,\alpha}(\bar f^-, \bar f^+) = \frac{1}{2}\left[F_{i,\alpha}(\bar f^-) + F_{i,\alpha}(\bar f^+) + |\Lambda|(\bar f_i^- - \bar f_i^+)n_\alpha\right], \tag{19}$$

where $\Lambda = \max(\frac{\partial F_{i,\alpha}}{\partial \bar f_i}n_\alpha) = c_{i,\alpha}n_\alpha$. Thus for the right hand side integrand of Eq. (18) we have:

$$[F_{i,\alpha}(\bar f) - F^*_{i,\alpha}(\bar f)]n_\alpha = \frac{1}{2}(c_{i,\alpha}n_\alpha - |c_{i,\alpha}n_\alpha|)(\bar f_i^- - \bar f_i^+). \tag{20}$$

In the next step of the nodal discontinuous Galerkin method, the field variables $\bar f_i$ are to be locally approximated in each element Ω^k as

$$\bar f_i^k(x_\alpha, t) \approx \hat f_i^k(x_\alpha, t) = \sum_{n=1}^{N_p}\hat f_{i,n}^k(t)\psi_n(x_\alpha) = \sum_{n=1}^{N_p}\hat f_i^k(x_{\alpha,n}, t)\ell_n^k(x_\alpha). \tag{21}$$

Here, the local approximate solution $\hat f_i^k(x_\alpha, t)$ is expanded in both the modal form with a local polynomial basis $\psi_n(x_\alpha)$ of order N, and the nodal representation, with the associated interpolating Lagrange polynomial, $\ell_n^k(x_\alpha)$. The connection between these two forms is through the definition of the expansion coefficients, $\hat f_{i,n}^k(t)$. $\hat f_i^k(x_{\alpha,n}, t)$ are the nodal values of the field variables and N_p is the number of the nodes in the element Ω^k. In this study we use triangular elements; Thus N_p is related to the order N of the basis polynomial as

$$N_p = (N+1)(N+2)/2 \tag{22}$$

As sketched in Fig. 1, a mapping $\mathbf{\Psi}$, is introduced connecting the general triangle with the standard triangle defined as $\mathrm{I} = \{\mathbf{r} = (r,s)\,|\,r,s \geqslant -1; r+s \leqslant 0\}$, that is [19]

$$\mathbf{x} = (x,y) = \mathbf{\Psi}(\mathbf{r}) = -\frac{r+s}{2}\mathbf{v}^1 + \frac{r+1}{2}\mathbf{v}^2 + \frac{s+1}{2}\mathbf{v}^3, \tag{23}$$

where $\mathbf{v}^1, \mathbf{v}^2$, and $\mathbf{v}^3$ are the vectors of element vertices counted counterclockwise. From Eq. (23) we have

$$\mathbf{x}_r = (x_r, y_r) = \frac{\mathbf{v}^2 - \mathbf{v}^1}{2}, \quad \mathbf{x}_s = (x_s, y_s) = \frac{\mathbf{v}^3 - \mathbf{v}^1}{2}. \tag{24}$$

However, we need the inverse derivatives in our relations, so using the relation

$$\frac{\partial \mathbf{x}}{\partial \mathbf{r}}\frac{\partial \mathbf{r}}{\partial \mathbf{x}} = \begin{bmatrix} x_r & x_s \\ y_r & y_s \end{bmatrix}\begin{bmatrix} r_x & r_y \\ s_x & s_y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \tag{25}$$

we obtain

$$r_x = \frac{y_s}{J}, \quad r_y = -\frac{x_s}{J}, \quad s_x = -\frac{y_r}{J}, \quad s_y = \frac{x_r}{J}, \tag{26}$$

where J is the Jacobian

$$J = x_r y_s - x_s y_r. \tag{27}$$

Through this mapping, we rewrite Eq. (21) in the (r,s) coordinates as

$$\hat f_i(\mathbf{r}, t) = \overset{N_p}{\underset{n=1}{m}}\hat f_{i,n}(t)\psi_n(\mathbf{r}) = \sum_{n=1}^{N_p}\hat f_i(\mathbf{r}_n, t)\ell_n(\mathbf{r}). \tag{28}$$

For the nodal discontinuous Galerkin method, Hesthaven and Warburton [19] used the orthonormal basis

$$\psi_m(\mathbf{r}) = \sqrt{2}P_i(a)P_j^{(2i+1,0)}(b)(1-b)^i, \quad i,j \geqslant 0, \quad i+j \leqslant N, \tag{29}$$

where

$$m = j + (N+1)i + 1 - i(i-1)/2, \quad a = 2(1+r)/(1-s) - 1, \quad b = s, \tag{30}$$

and $P_j^{(\alpha,\beta)}(x)$ is the j-th order Jacobi polynomial.

In using the interpolation Eq. (39), we are left with identifying N_p points on the standard triangle I, leading to well-behaved interpolations. This choice is important since a poorly chosen set can result in computational problems due to ill-conditioning [19]. We

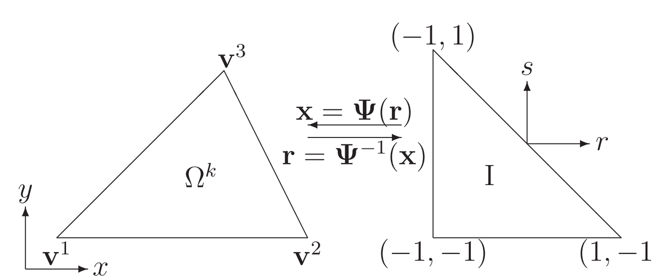

Fig. 1. The mapping $\mathbf{\Psi}$ maps the standard triangle I into the general triangle Ω^k.

employ the method proposed by Hesthaven and Warburton [19] to locate N_p points on the equilateral triangle and then map the computed nodes into the standard triangle I.

Now, we substitute the interpolation Eq. (39) into the local weak form Eq. (18), with the classic Galerkin choice for the test function, which requires that the spaces spanned by the basis functions and the test functions are the same. This leads to the following space-discretized equation:

$$\mathbf{M}^k \frac{d\mathbf{f}_i^k}{dt} + \mathbf{S}_i^k \mathbf{f}_i^k = \mathbf{R}_i^k, \tag{31}$$

where $\mathbf{f}_i^k = [\hat{f}_i^k]_n$, $(n = 1, \ldots, N_p)$, is the solution vector, and the components of the element mass matrix $\mathbf{M}^k$, stiffness matrix $\mathbf{S}_i^k$, and right hand side matrix $\mathbf{R}_i^k$, are

$$[\mathbf{M}^k]_{mn} = \int_{\mathrm{I}} \ell_m(\mathbf{r}) \ell_n(\mathbf{r}) J^k d\mathbf{r}, \quad (m, n = 1, \ldots, N_p), \tag{32}$$

$$[\mathbf{S}_i^k]_{mn} = \int_{\mathrm{I}} \mathbf{c}_i \cdot \frac{\partial \ell_m(\mathbf{r})}{\partial \mathbf{r}} \frac{\partial \mathbf{r}}{\partial \mathbf{x}} \ell_n(\mathbf{r}) J^k d\mathbf{r}, \quad (m, n = 1, \ldots, N_p), \tag{33}$$

$$[\mathbf{R}_i^k]_{mn} = \int_{\partial \mathrm{I}} \frac{1}{2} (\mathbf{c}_i \cdot \mathbf{n} - |\mathbf{c}_i \cdot \mathbf{n}|)(\hat{f}_i^- - \hat{f}_i^+)_m \ell_m(\mathbf{r}) \ell_n(\mathbf{r}) J_s^k d\mathbf{r},$$
$$(m, n = 1, \ldots, N_p, \ m, n \text{ are on } \partial \mathrm{I}), \tag{34}$$

where J^k and J_s^k are the element surface and perimeter mapping Jacobians, respectively, and $\partial \mathrm{I}$ is the perimeter of the standard triangle I.

2.3. Runge–Kutta time marching

For the time integration of the semi-discrete DG Eq. (31), we make use of an explicit Runge–Kutta method. To do so, we first multiply both sides of Eq. (31) by the inverse of the matrix $\mathbf{M}^k$ to arrive at

$$\frac{d\mathbf{f}_i^k}{dt} = -\mathbf{M}^{k^{-1}} \mathbf{S}_i^k \mathbf{f}_i^k + \mathbf{M}^{k^{-1}} \mathbf{R}_i^k = \mathcal{L}(\mathbf{f}_i^k). \tag{35}$$

We apply the low-storage fourth-order, five-stage Runge–Kutta method (LSERK) of the form

$$\mathbf{p}^{(0)} = \mathbf{f}_i^{k^{(n)}},$$
$$j = 1, \ldots, 5 : \begin{cases} \mathbf{k}^{(j)} = a_j \mathbf{k}^{(j-1)} + \delta t \mathcal{L}(\mathbf{p}^{(j-1)}, t^{(n)} + c_j \delta t), \\ \mathbf{p}^{(j)} = \mathbf{p}^{(j-1)} + b_j \mathbf{k}^{(j)}, \end{cases} \tag{36}$$
$$\mathbf{f}_i^{k^{(n+1)}} = \mathbf{p}^{(5)},$$

where a_j, b_j, and c_j are given constants [19].

2.4. Boundary conditions

In the fluid flow cases to be considered in this study, we have only inlet or outlet and wall boundaries. These boundary conditions are weakly imposed through the right hand side matrix i.e. Eq. (34). That is, for the boundary unknown particle distribution $(\hat{f}_i^+)$ we set

- inlet and outlet:

$$\hat{f}_i^+ = \hat{f}_i^-, \quad \mathbf{e}_i \cdot \mathbf{n} > 0 \, (\text{outgoing particle}),$$
$$\hat{f}_i^+ = f_i^{\mathrm{eq}}(\rho_0, \mathbf{u}_0), \quad \mathbf{e}_i \cdot \mathbf{n} < 0 \, (\text{incoming particle}), \tag{37}$$

- wall:

$$\hat{f}_i^+ = \hat{f}_i^-, \quad \mathbf{e}_i \cdot \mathbf{n} > 0 \, (\text{outgoing particle}),$$
$$\hat{f}_i^+ = \hat{f}_{i*}^- + 2 t_i \rho_0 (\mathbf{e}_i \cdot \mathbf{u}_0)/c_s^2, \quad \mathbf{e}_i \cdot \mathbf{n} < 0 \, (\text{incoming particle}), \tag{38}$$

where ρ_0 and $\mathbf{u}_0$ are the macroscopic values of density and velocity at the boundary, respectively, and $\hat{f}_{i*}^-$ is the distribution function of the particle moving in the opposite direction of $\hat{f}_i^-$. It should be noted that in Eq. (38), the incoming particle distribution for the wall boundary is derived by equating non-equilibrium distribution functions of incoming and outgoing particles (no-slip condition).

3. Results and discussions

To validate the numerical method, two example problems are simulated; the lid driven square cavity at Reynolds numbers of 1000 and 5000 and a Mach number of 0.1, and the impulsively started cylinder at Reynolds numbers of 550 (Mach number of 0.1) and 9500 (Mach number of 0.05). Next, the method is used to evaluate the permeability of a porous medium.

3.1. The lid-driven square cavity

The lid driven square cavity is a well-known benchmark problem for which both numerical and experimental data is available.

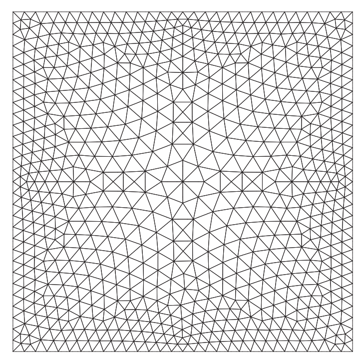

Fig. 2. Unstructured grid made of 1440 triangular elements for the lid-driven cavity flow.

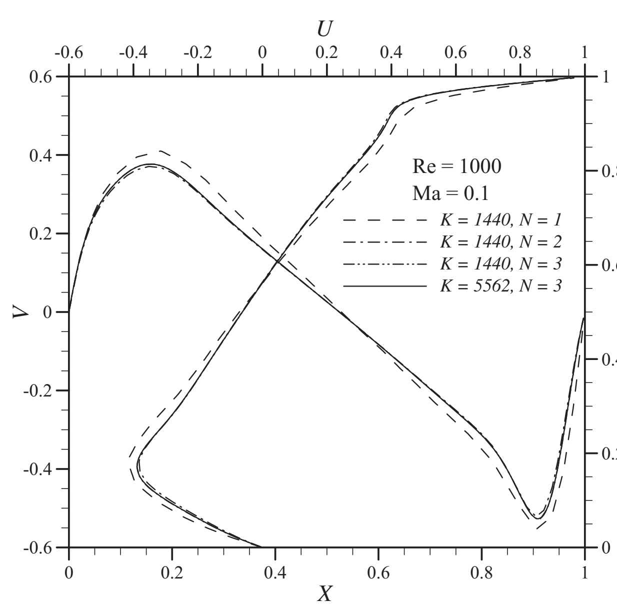

Fig. 3. Convergence of the centerline velocities at Reynolds number of 1000 and Mach number of 0.1 for different number of elements K and different order of interpolating polynomials N.

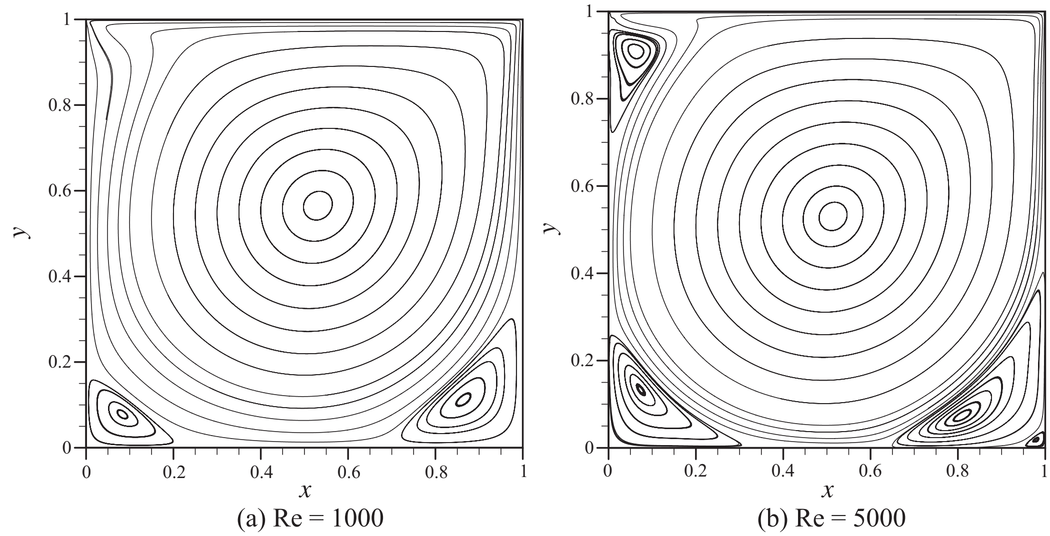

Fig. 4. Streamlines of the lid-driven cavity flow at Reynolds numbers of (a) 1000 and (b) 5000.

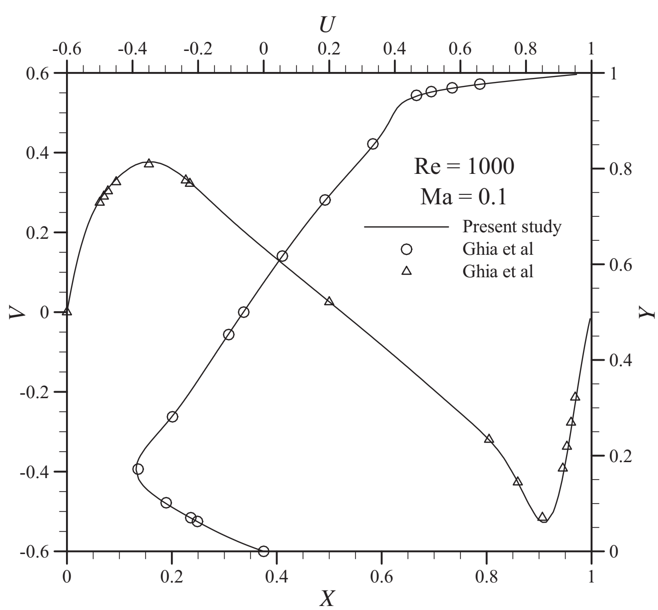

Fig. 5. Centerline velocity profiles at Re = 1000 and Ma = 0.1 obtained from the present study (lines) compared with results reported by Ghia et al. (symbols).

Our objective here is to simulate flows at moderate to high Reynolds numbers and to study the accuracy of the results.

We began our simulations with an unstructured grid made of $K = 1440$ triangular elements (Fig. 2) and used first order interpolating polynomials ($N = 1$). As the order of the interpolating polynomial (N) is increased, the number of nodes per element (N_p) increases according to Eq. (22). We then applied second and third order interpolating polynomials, respectively. To confirm the convergence of the results, we also used a very fine grid with 5562 elements and applied the third order interpolating polynomials. The dimensionless velocity profiles U and V at the centerlines of the cavity are illustrated in Fig. 3 for the two grids and different order of interpolating polynomials N. It can be seen that the results converge as N increases or equivalently as the grid itself is refined.

Fig. 4 shows the Streamlines of the lid-driven cavity flow at Reynolds numbers of 1000 and 5000. For Re = 1000, three vortices namely the main (central), the lower left, and the lower right vortices are observed. For Re = 5000, the upper left vortex appears in the flow, as it is expected.

Table 1
Coordinates of the centers of the central, lower left, and lower right vortices obtained from this work and from the works of others.

Re	Ref	Central		Lower left		Lower right	
		X	Y	X	Y	X	Y
1000	present	0.5308	0.5653	0.0832	0.0780	0.864	0.1118
	Hou	0.5333	0.5647	0.0902	0.0784	0.8667	0.1137
	Ghia	0.5313	0.5625	0.0859	0.0781	0.8594	0.1094
5000	present	0.5142	0.5305	0.0736	0.1330	0.8096	0.0732
	Hou	0.5176	0.5373	0.0784	0.1373	0.8078	0.0745
	Ghia	0.5117	0.5352	0.0703	0.1367	0.8076	0.0742

For the verification of the results of our simulation, the centerline velocity profiles at the Reynolds number of 1000 and the Mach number of 0.1 are compared with the results reported by Ghia et al. [28] in Fig. 5. It can be seen that they are in excellent agreement. Moreover, in Table 1, the coordinates of the centers of the central, lower left, and lower right vortices, which are obtained from the present work, are compared with the results reported by Ghia et al. [28] and Hou et al. [29]. It can be concluded from the table that the results for the main (central) vortex are in excellent agreement with the previous results in the literature (the maximum difference is about 1%). For the lower left and lower right vortices, the maximum difference between our results and those of Ghia et all is about 4% and between our results and those of Hou et all is about 7%.

3.2. Impulsively started cylinder

As the second numerical test, we simulate the impulsively started transient flow around a cylinder at Reynolds numbers of 550 (Mach number of 0.1) and 9500 (Mach number of 0.05). The space has been discretized using 13,135 unstructured triangular elements, as shown in Fig. 6. To capture the boundary layer features of the flow accurately, the grid becomes finer around the cylinder in a systematic way shown in the figure. In Fig. 6(e) we show the computational nodes on the grid when $N = 3$. We assume a potential flow at $t = 0$ to suppress the numerical effects due to initially developed thin boundary layers associated with the impulsive start. At $t > 0$ the no-slip boundary condition is imposed on

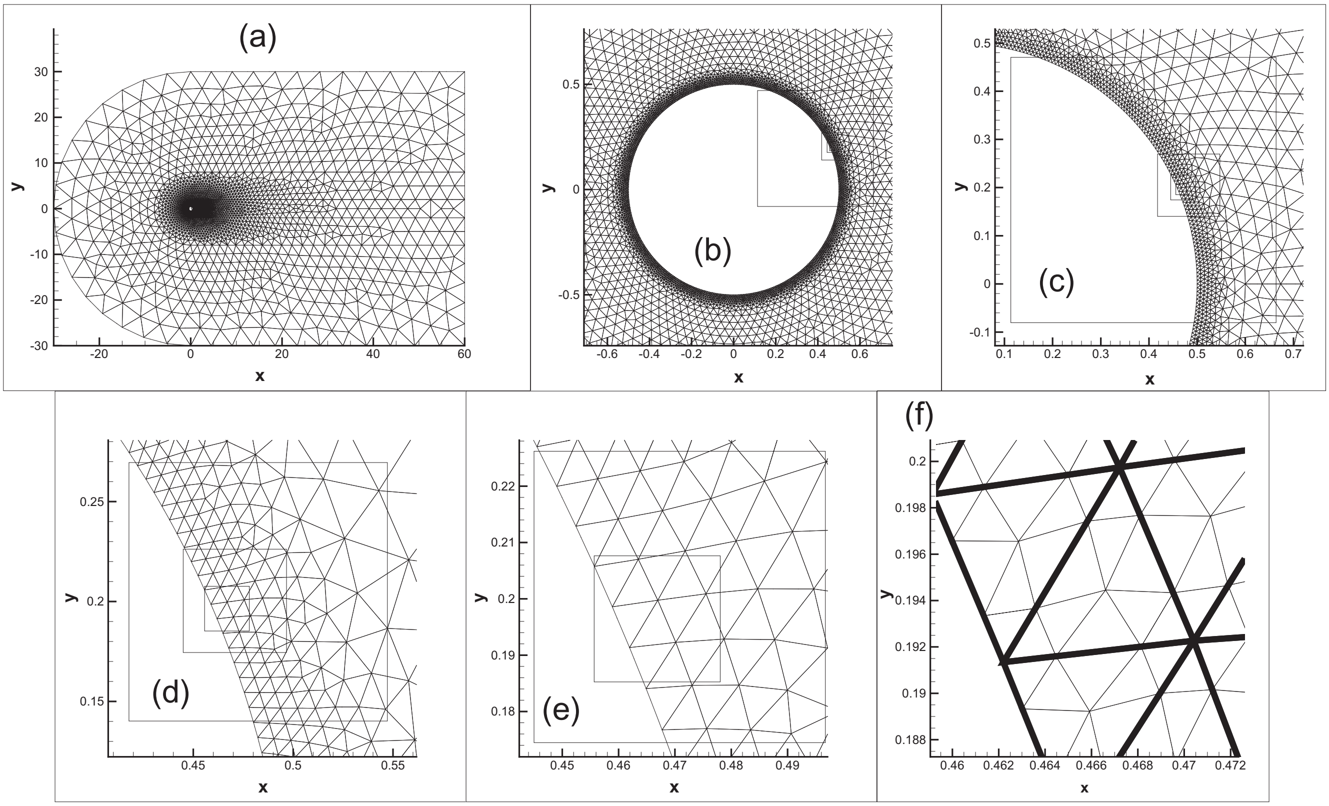

Fig. 6. The computational domain and the unstructured grid made of 13,135 triangular elements for the impulsively started cylinder flow. In subfigure (f) the nodes inside each element are shown for $N = 3$.

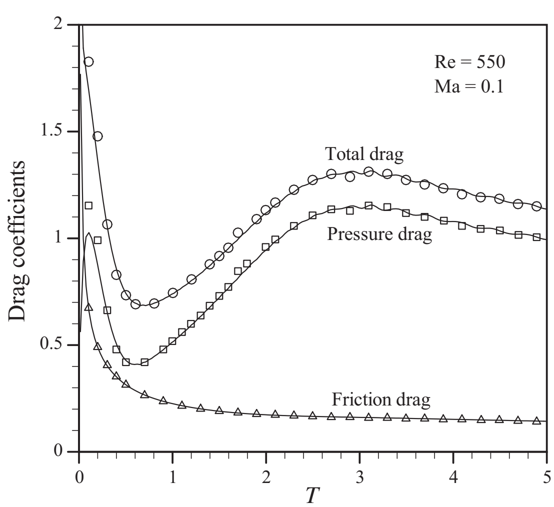

Fig. 7. Comparison between the friction, pressure, and total drag coefficients obtained from the present study (lines) and reported by Min and Lee [24] (symbols) at Reynolds number of 550 and Mach number of 0.1.

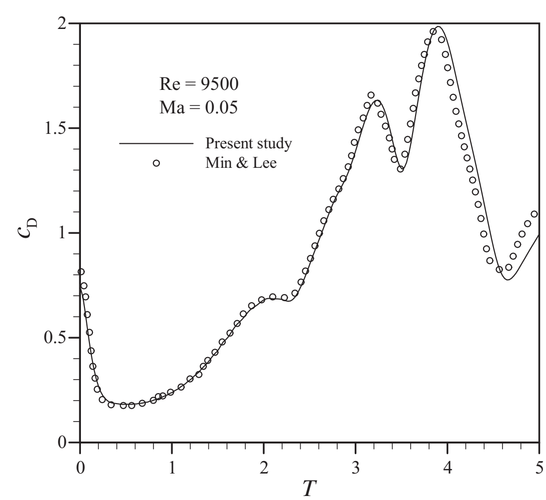

Fig. 8. Comparison between the total drag coefficients obtained from the present study (line) and reported by Min and Lee [24] (symbols) at Reynolds number of 9500 and Mach number of 0.05.

the surface of the cylinder and constant density and velocity is imposed for all other boundaries.

To validate the results, we compare our results with the results reported by Min and Lee [24]. Fig. 7 shows the friction, pressure, and total drag coefficients with respect to the dimensionless time, obtained from our simulation and those of Min and Lee for the Reynolds number of 550 and the Mach number of 0.1. Excellent agreement between the two results is observed. The total drag coefficient (which is mainly influenced by the pressure field) with respect to the dimensionless time for the Reynolds number of 9500 and the Mach number of 0.05 is depicted in Fig. 8. The results of the present study are in good agreement with those of Min and Lee.

Figs. 9 and 10 show the streamlines at various dimensionless times of the current simulation for the Re = 550, Ma = 0.1 and Re = 9500, Ma = 0.05 cases, respectively. The flow separation and the development of the symmetric vortices behind the cylinder are observed in these figures.

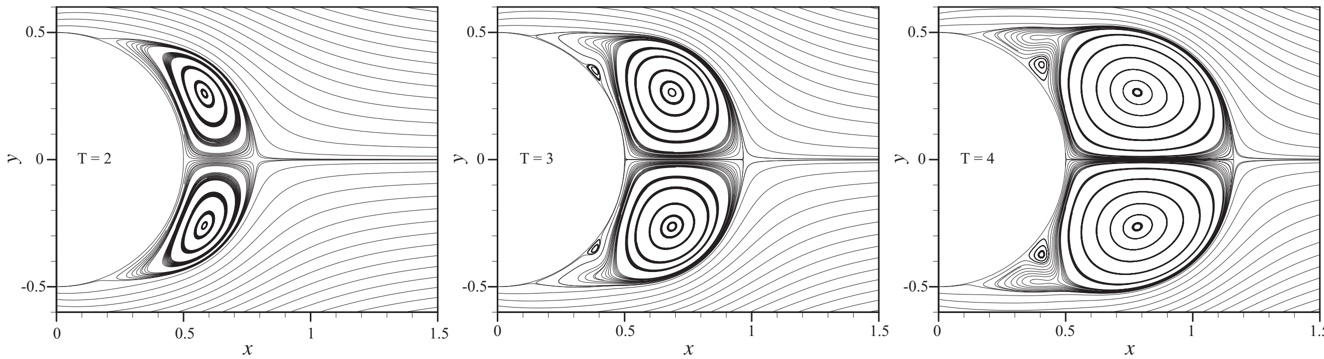

Fig. 9. Streamlines at dimensionless time T = 2, 3, 4 for Re = 550 and Ma = 0.1.

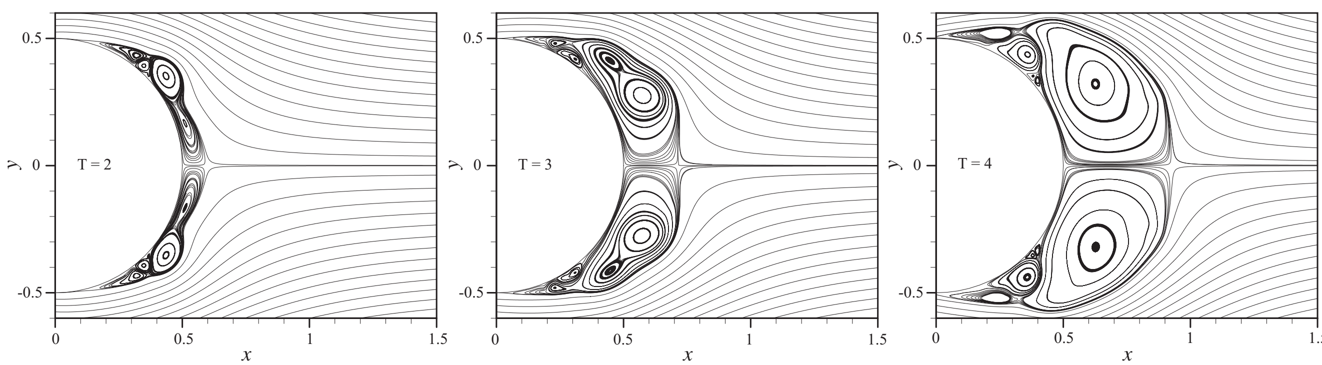

Fig. 10. Streamlines at dimensionless time T = 2, 3, 4 for Re = 9500 and Ma = 0.05.

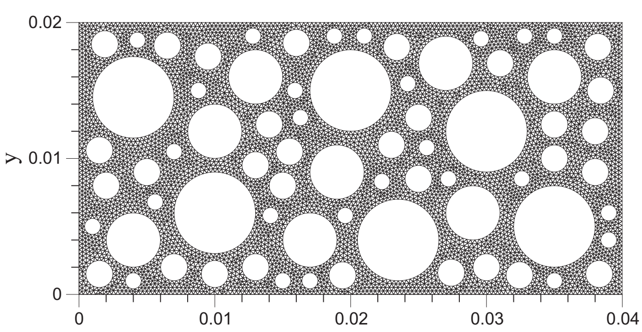

Fig. 11. The domain and the unstructured mesh with triangular elements for the flow in a porous medium.

3.3. Flow in a porous medium

One of the most common applications of the standard lattice Boltzmann method is the simulation of flow in porous media. In order to illustrate the ability of the nodal discontinuous Galerkin lattice Boltzmann method (NDG-LBM) to deal with complex geometries, we simulate the pressure-driven fluid flow in a typical two-dimensional porous medium shown in Fig. 11 and compute the permeability of the medium. One of the common definitions of the Reynolds number for the porous media flow is

$$Re = \frac{\rho v D_p}{\mu}, \tag{39}$$

where ρ and μ are the density and viscosity of the fluid, v is the superficial flow velocity, and D_p is a representative grain size for

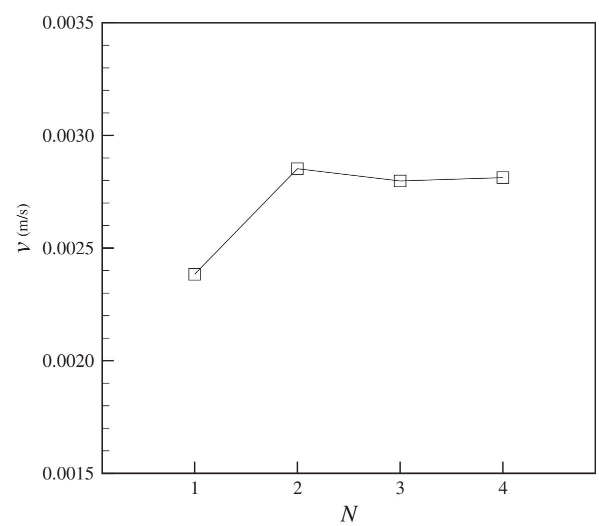

Fig. 12. The superficial velocity versus the order of the basis polynomials.

the porous media. Experimental studies have illustrated that flow regimes with low Reynolds numbers (typically less than 10) are Darcian, meaning that the superficial velocity–pressure gradient relation obeys the Darcy's law

$$v = -\frac{k}{\mu}\nabla p, \tag{40}$$

where k is the permeability of the medium measured in m^2 or *Darcy*, where 1 Darcy $= 9.869233 \times 10^{-13}$ m^2.

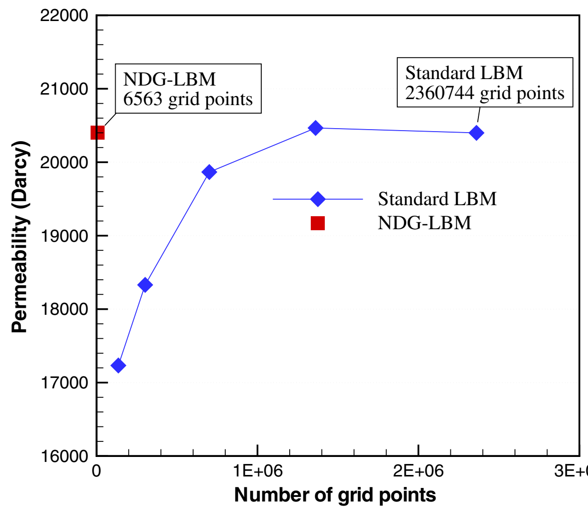

Fig. 13. The permeability of the porous medium versus the number of grid points, a comparison between NDG-LBM and the standard LBM.

In this study, simulations are performed on a 4 cm $\times$ 2 cm specimen with a porosity of 0.5 as shown in Fig. 11. The no-slip boundary conditions are imposed on the top and the bottom walls and on the grain surfaces through the bounce-back scheme of Eq. (23) and the constant pressure boundary condition on the left and right boundaries are imposed through Eq. (22). The pressure difference between inlet and outlet boundaries in our simulations are set to $\nabla p = 2.5\ \mathrm{Pa/m}$, so that the Reynolds number of the flow is less than 1, which guaranties the Darcian flow regime.

The domain is discretized using 11,100 triangular elements, equivalent to 6563 grid points, as shown in Fig. 11. First, to investigate the convergence of the numerical method with respect to the domain discretization, we set the order of the basis polynomials as $N = 1, 2, 3$, and 4 and compute the superficial velocity in all the cases. The results are depicted in Fig. 12. As shown in this figure, the differences between the cases with $N = 2$, $N = 3$, and $N = 4$ is less than 1.5% and therefore the order of the polynomial basis $N = 2$ is sufficient for our simulations.

Next, to compare our method with the standard LBM, we solve the flow in the porous medium using the standard LBM. Fig. 13 illustrates the results for the permeability of the domain computed using NDG-LBM and the standard LBM. It is observed from this figure that the permeability of the domain computed from the standard LBM converges to the result of NDG-LBM at a significantly greater number of gird points.

4. Conclusion

We present a new application of the nodal discontinuous Galerkin method in solving the lattice Boltzmann equation. By applying the operator-splitting in the lattice Boltzmann equation and using the unstructured triangular mesh of the nodal discontinuous Galerkin method for discretizing the streaming equation, our method has been capable of simulating well known fluid flow benchmark problems with numerical stability at high Reynolds numbers. Although the element mass matrix in our method is not diagonal, this issue in not as important since the element mass matrix is independent of time, therefore the inverse of the element mass matrices is calculated only at the beginning of the computations.

In order to validate the present method and the implemented computer code, two well-known test cases, namely the lid-driven cavity flow at Reynolds numbers of 1000 and 5000 and a Mach number of 0.1, and the impulsively started cylinder flow at Reynolds numbers of 550 (Ma = 0.1) and 9500 (Ma = 0.05) were simulated in this study. The results show excellent agreement with other benchmark results in the literature. The method is then applied for estimating the permeability of a porous medium. It was illustrated that the required number of grid points of NDG-LBM for simulating a geometrically complex flow, i.e. the flow in the porous medium in this study, is significantly lower than that of the standard LBM.

References

[1] Benzi R, Succi S, Vergassola M. The lattice Boltzmann equation: theory and applications. Phys Rep 1992;222(3):145–97.
[2] Chen S, Doolen GD. Lattice Boltzmann method for fluid flows. Annu Rev Fluid Mech 1998;30(1):329–64.
[3] McNamara GR, Zanetti G. Use of the Boltzmann equation to simulate lattice-gas automata. Phys Rev Lett 1988;61(20):2332–5.
[4] McNamara GR, Garcia AL, Alder BJ. Stabilization of thermal lattice Boltzmann models. J Stat Phys 1995;81(1–2):395–408.
[5] Sterling JD, Chen S. Stability analysis of lattice Boltzmann methods. J Comput Phys 1996;123(1):196–206.
[6] He X, Luo L-S. Theory of the lattice Boltzmann method: from the Boltzmann equation to the lattice Boltzmann equation. Phys Rev E 1997;56(6):6811.
[7] He X, Luo L-S, Dembo M. Some progress in lattice Boltzmann method. Part i. Nonuniform mesh grids. J Comput Phys 1996;129(2):357–63.
[8] Cao N, Chen S, Jin S, Martinez D. Physical symmetry and lattice symmetry in the lattice Boltzmann method. Phys Rev E 1997;55(1):R21–4.
[9] Mei R, Shyy W. On the finite difference-based lattice Boltzmann method in curvilinear coordinates. J Comput Phys 1998;143(2):426–48.
[10] Sofonea V, Sekerka RF. Viscosity of finite difference lattice Boltzmann models. J Comput Phys 2003;184(2):422–34.
[11] Bardow A, Karlin I, Gusev A. General characteristic-based algorithm for off-lattice Boltzmann simulations. EPL (Europhys Lett) 2006;75(3):434.
[12] Xi H, Peng G, Chou S-H. Finite-volume lattice Boltzmann method. Phys Rev E 1999;59(5):6202.
[13] Ubertini S, Bella G, Succi S. Lattice Boltzmann method on unstructured grids: further developments. Phys Rev E 2003;68(1):016701.
[14] Stiebler M, Tölke J, Krafczyk M. An upwind discretization scheme for the finite volume lattice Boltzmann method. Comput Fluids 2006;35(8):814–9.
[15] Patil DV, Lakshmisha K. Finite volume tvd formulation of lattice Boltzmann simulation on unstructured mesh. J Comput Phys 2009;228(14):5262–79.
[16] Lee T, Lin C-L. A characteristic Galerkin method for discrete Boltzmann equation. J Comput Phys 2001;171(1):336–56.
[17] Lee T, Lin C-L. An Eulerian description of the streaming process in the lattice Boltzmann equation. J Comput Phys 2003;185(2):445–71.
[18] Li Y, LeBoeuf EJ, Basu P. Least-squares finite-element scheme for the lattice Boltzmann method on an unstructured mesh. Phys Rev E 2005;72(4):046711.
[19] Hesthaven JS, Warburton T. Nodal discontinuous Galerkin methods: algorithms, analysis, and applications, vol. 54. New York: Springer-Verlag; 2008.
[20] Shi X, Lin J, Yu Z. Discontinuous Galerkin spectral element lattice Boltzmann method on triangular element. Int J Numer Methods Fluids 2003;42(11):1249–61.
[21] Dubiner M. Spectral methods on triangles and other domains. J Sci Comput 1991;6(4):345–90.
[22] Düster A, Demkowicz L, Rank E. High-order finite elements applied to the discrete Boltzmann equation. Int J Numer Methods Eng 2006;67(8):1094–121.
[23] Szabo B, Babuška I. Finite element analysis. Wiley-Interscience; 1991.
[24] Min M, Lee T. A spectral-element discontinuous Galerkin lattice Boltzmann method for nearly incompressible flows. J Comput Phys 2011;230(1):245–59.
[25] Bhatnagar PL, Gross EP, Krook M. A model for collision processes in gases. i. Small amplitude processes in charged and neutral one-component systems. Phys Rev 1954;94(3):511.
[26] He X, Chen S, Doolen GD. A novel thermal model for the lattice Boltzmann method in incompressible limit. J Comput Phys 1998;146(1):282–300.
[27] Wolf-Gladrow DA. Lattice-gas cellular automata and lattice Boltzmann models: an introduction, vol. 1725. Springer-Verlag; 2000.
[28] Ghia U, Ghia K, Shin C. High-re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method. J Comput Phys 1982;48(3):387–411.
[29] Hou S, Zou Q, Chen S, Doolen G, Cogley AC. Simulation of cavity flow by the lattice Boltzmann method. J Comput Phys 1995;118(2):329–47.